

DAU Dashboard Architecture & CCU Analysis

Sandbox Data Engineer — Take-home presentation

Business Intelligence in Games • DAU Dashboard Design • CCU Query

Target scale

30M accounts • 500M sessions

Business Intelligence in Games

What data is produced, who uses it, and which KPIs support decisions

DATA SOURCES

- Accounts and player profiles
- Sessions and gameplay events
- Payments and monetization
- Marketing / campaign data
- Technical telemetry

The task focuses on accounts + session data.

STAKEHOLDERS

- Product
- Game design / live ops
- Marketing
- Finance
- Engineering / operations

Different teams ask different questions.

TYPICAL KPIs

- DAU / MAU
- Retention
- CCU / concurrency
- Revenue / ARPU
- Feature / player engagement

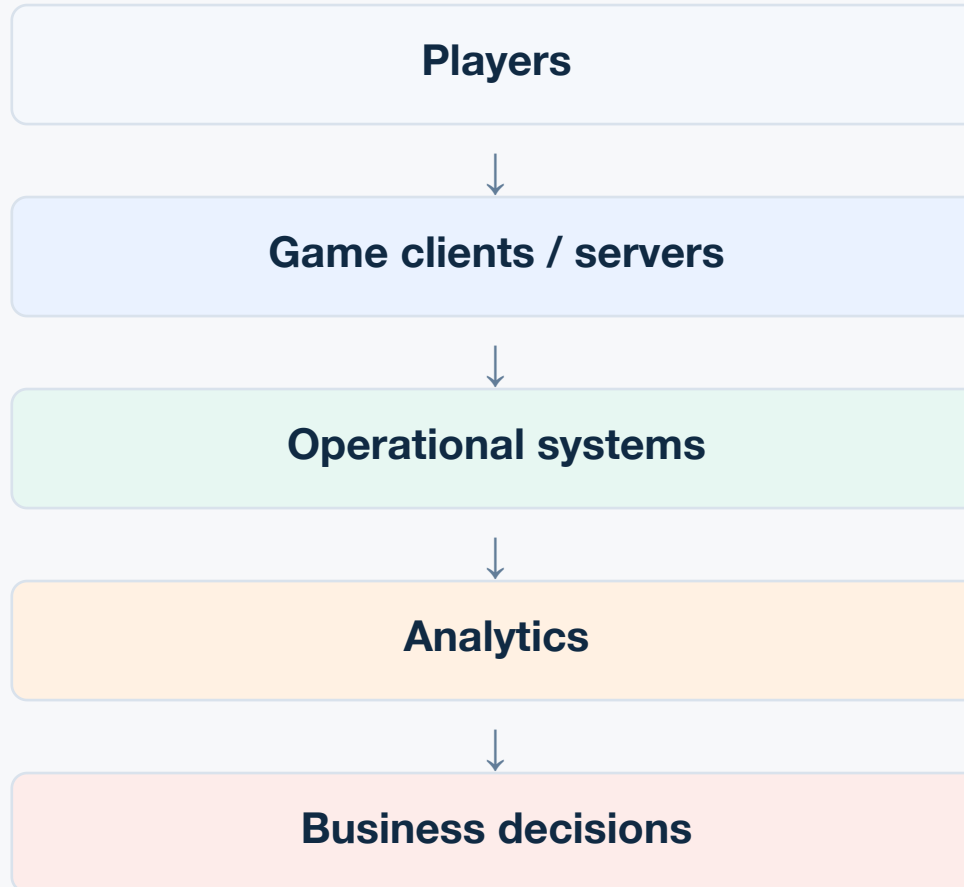
The assignment asks us to turn DAU into a dashboard.

GaaS data flow

From player behavior to business decisions

Typical data domains

- Player
- Gameplay
- Technical
- Monetization
- Marketing



Why the data layer matters

- Combine sources
- Define metrics
- Make results reusable
- Support decisions

This task focuses on DAU as one analytical decision layer.

Requirements

Translate the task into engineering constraints

30M

accounts

500M

sessions

Unique

DAU

Multi-year

date range

Country

filter

Platform

filter

Looker Studio

dashboard

The design implication

Separate source systems from analytics, process incrementally, model DAU at a reusable grain, and keep dashboard queries small.

The source data is explicitly PostgreSQL accounts + Elasticsearch sessions.

Architecture

Keep operational sources separate from the analytical layer



The local prototype uses DuckDB as the BigQuery analogue.

Data model

Choose a grain that preserves the dimensions the dashboard needs

fact_daily_user_activity

activity_date

date / partition

user_id

user

country

dashboard filter

platform

dashboard filter

Grain: 1 row per user / day / platform

Why this grain?

Platform stays in the grain because a user can be active on multiple platforms on the same day.

Country is carried into the fact table because it is a common dashboard filter.

DAU calculation & scaling

Make the metric correct first, then make it cheap to query

DAU = unique users active on a calendar day

```
SELECT activity_date, COUNT(DISTINCT
user_id) AS dau
FROM fact_daily_user_activity
GROUP BY activity_date;
```

Why DISTINCT?

- Multiple sessions should count as one user.
- Multiple platforms still represent one overall user.
- Overall DAU must not sum platform DAU.

Scaling choices

Incremental loads

Process new / changed windows, not full history

Late-event reprocessing

Reprocess a bounded recent window

Idempotent MERGE

Same logical row can be safely rewritten

Partition: activity_date

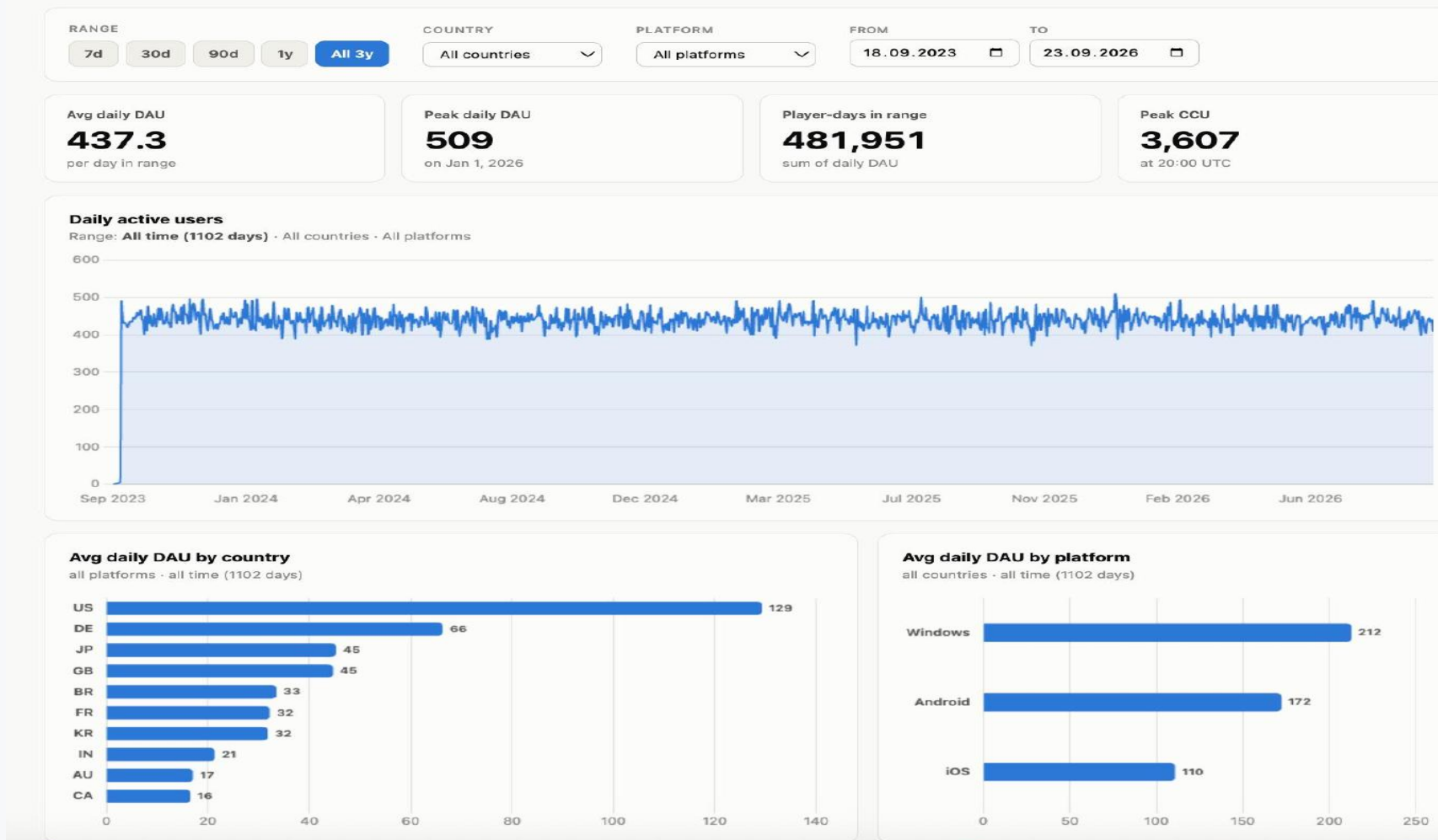
Date filters scan fewer partitions

Cluster: country, platform

Common dashboard filters stay efficient

Dashboard

A small analytical model powers reusable visual analysis



What it shows

- DAU trend over time
- Country comparison
- Platform comparison
- Date filters
- Country / platform filters

Prototype: local data + public web dashboard
Production target: BigQuery + Looker Studio

CCU – Concurrent Users

Turn minute-level concurrency into an hourly metric

Input: minute-level CCU

10:00	120
10:01	125
10:02	131
...	...
10:59	—

Hourly result

AVG(minute-level CCU)

Submitted interpretation

“CCU per hour” = average of the minute-level CCU values within that hour.

Business-definition caveat

If the business actually means peak hourly concurrency, use MAX(ccu) instead of AVG(ccu).

The important thing is to make the metric definition explicit.

Trade-offs & design choices

Use the simplest architecture that satisfies the workload

Why BigQuery?

Analytical workload, large historical volume, SQL + BI integration, managed infrastructure.

Why no Spark?

The problem is primarily analytical SQL; BigQuery already provides distributed processing, so Spark adds operational complexity without a demonstrated need.

Why not query Elasticsearch?

Keep dashboard workloads separate from the operational/event source system.

Why daily activity?

Pre-shape the user/day/platform result so the dashboard does not repeat expensive raw-session analysis.

Principle: separate operational systems, transformations, analytical storage, and BI consumption.

Thank you

DAU dashboard • scalable analytical model • explicit metric definitions

Questions